

Property Calculation Methods

Supplementary note for plot_h2_ch4_co2_properties.py

This document describes how each thermophysical property shown in Figures 1-3 is computed.

Gases considered: H₂, CH₄, CO₂. Temperatures: 298.15, 323.15, 373.15 K. Pressures: 1-400 bar (Figs

1. Density (Figure 1a)

Source: CoolProp library (<http://www.coolprop.org>)

Density is obtained directly from the CoolProp implementation of multi-parameter Helmholtz free-energy equations of state. The specific EOS used for each fluid is:

H₂ Leachman et al. (2009), J. Phys. Chem. Ref. Data 38(3)
CH₄ Setzmann & Wagner (1991), J. Phys. Chem. Ref. Data 20(6)
CO₂ Span & Wagner (1996), J. Phys. Chem. Ref. Data 25(6)

The CoolProp call is:

$$\rho(T, P) = \text{PropsSI}('D', 'T', T [\text{K}], 'P', P [\text{Pa}], \text{fluid})$$

where 'D' requests mass density in kg/m³. Invalid or two-phase state points are caught by a try/except block and replaced with NaN.

2. Isothermal Compressibility (Figure 1b)

Definition:

The isothermal compressibility is defined as the fractional change in density with pressure at constant temperature. It is NOT the compressibility factor $Z = Pv/(RT)$.

$$\beta_T = \frac{1}{\rho} \left(\frac{\partial \rho}{\partial P} \right)_T \quad [\text{Pa}^{-1}]$$

Numerical method:

The partial derivative is approximated with a second-order central finite difference:

$$\left(\frac{\partial \rho}{\partial P} \right)_T \approx \frac{\rho(T, P + \Delta P) - \rho(T, P - \Delta P)}{2 \Delta P}$$

3. Gas-Water Interfacial Tension (Figure 2a)

Source:

IFT values are hard-coded literature data (digitised from published experimental curves). No model or correlation is applied; the data points are plotted as-is with linear interpolation between markers.

The data is stored as Python dictionaries keyed by gas name and temperature (integer K). Each entry contains arrays of pressure [MPa] and IFT [mN/m].

Temperatures in the IFT table use integer keys (298, 323, 373) mapped from the floating-point values 298.15, 323.15, 373.15 K used elsewhere in the script.

Key physical observations:

- H₂-water IFT decreases only slightly with pressure (~72 to ~69 mN/m over 0-45 MPa).
- CH₄-water IFT shows moderate reduction (~72 to ~52 mN/m).
- CO₂-water IFT drops dramatically near the CO₂ phase transition (~72 to ~23 mN/m), reflecting the much stronger CO₂-water molecular interaction and near-critical density changes.

4. Approximate Solubility (Figure 2b)

Model:

Dissolved-gas mole fraction in pure water is estimated using Henry's law:

$$x = \frac{P}{H(T)}$$

where P is the total pressure [MPa] and H(T) is the Henry-law constant [MPa] at temperature T.

Henry-law constants used (MPa):

	298.15 K	323.15 K	373.15 K
H ₂	7100	7500	6900
CH ₄	4000	4600	5800
CO ₂	167	260	440

5. Dynamic Viscosity (Figure 3a)

Source:

Viscosity is obtained from CoolProp using the same Helmholtz-based EOS framework. CoolProp internally uses the following transport-property correlations:

H2 Muzny et al. (2013), J. Chem. Eng. Data 58(4)
CH4 Quinones-Cisneros et al. (2012), J. Chem. Eng. Data 57(12)
CO2 Fenghour et al. (1998), J. Phys. Chem. Ref. Data 27(1)

The CoolProp call is:

$$\mu(T, P) = \text{PropsSI}('V', 'T', T [\text{K}], 'P', P [\text{Pa}], \text{fluid}) \quad [\text{Pa}\cdot\text{s}]$$

The result is converted to micro-Pascal-seconds for plotting:

$$\mu [\mu\text{Pa}\cdot\text{s}] = \mu [\text{Pa}\cdot\text{s}] \times 10^6$$

6. Approximate Binary Gas-Phase Diffusion Coefficient (Figure 3b)

Correlation:

The Fuller-Schettler-Giddings (1966) correlation for gas-phase binary diffusion:

$$D_{AB} [\text{cm}^2/\text{s}] = \frac{0.00143 T^{1.75} \sqrt{M_A^{-1} + M_B^{-1}}}{P (\sum V_A^{1/3} + \sum V_B^{1/3})^2}$$

where:

T = temperature [K]
P = pressure [bar]
MA, MB = molecular weights [g/mol]
Sv_A, Sv_B = Fuller atomic diffusion volumes [-]

7. Unit Conversions

All internal calculations are performed in SI units. The following conversions are applied for plotting:

Pressure: 1 bar = 1e5 Pa = 0.1 MPa
Density: [kg/m³] (no conversion)
Compressibility: 1 Pa⁻¹ = 1e6 MPa⁻¹
Viscosity: 1 Pa.s = 1e6 uPa.s
Diffusivity: [cm²/s] (Fuller output is already in cm²/s)
IFT: [mN/m] (literature data as-is)
Solubility: [-] (mole fraction, dimensionless)

8. Summary of Data Sources

Property	Source	Rigour
Density	CoolProp (Helmholtz EOS)	Reference-quality
Isothermal compressibility	Finite-diff. of CoolProp density	Reference-quality
Dynamic viscosity	CoolProp (transport corr.)	Reference-quality
Gas-water IFT	Tabulated literature data	Experimental
Solubility	Henry-law approximation	Screening only
Diffusion coefficient	Fuller gas-phase corr.	Screening only

9. Error Handling

All CoolProp queries are wrapped in a `safe_coolprop()` helper that catches any exception (e.g. two-phase input, out-of-range state) and returns NaN. NaN values produce gaps in the plotted curves rather than crashes. This is particularly relevant for CO₂ near its critical point ($T_c = 304.13$ K, $P_c = 73.77$ bar) and along the saturation curve at 298.15 K.